

# [DC] The Effect of Modulating The Step Length of an Embodied Self-Avatars on Gait Symmetry During Treadmill Walking

Iris Willaert  
Ecole de technologie  
superieure  
Montreal, Canada

## ABSTRACT

In virtual reality, it is possible to simulate visual self-representation by mapping one's body movements to those of an avatar. The acceptance of the virtual body as part of one's own body creates an ownership illusion. This paper discusses different protocols to study the effects of modulating a self-avatar's step length on the step length of embodied healthy participants and post-stroke patients.

**Keywords:** Self-avatar, virtual reality, ownership illusion, rehabilitation, visual feedback, gait symmetry.

**Index Terms:** H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, Augmented, and Virtual Realities; H.5.2 [Information Interfaces and Presentation]: User Interfaces—Evaluation/Methodology.

## 1 INTRODUCTION

Virtual reality (VR) has been accepted to be a useful tool for assessment and treatment in rehabilitation [1]. In immersive VR, virtual bodies (or self-avatars) that replicate the user's movements can be used to substitute the real body, which is not visible inside the virtual environment (VE). If the movements of the avatar are sufficiently synchronized with the user's actual movements, the user can experience a sense of ownership over the avatar. Previous studies have shown that altering an embodied avatars movements can lead to changes in a user's movement. The conflict between visual and proprioceptive signals led participants to unconsciously alter their own body movements towards those of the avatar in order to diminish the incongruency between the real and virtual bodies [2, 3]. However, to date, not much is known about the physical limits of these unconscious motor adaptations. Understanding the perception threshold between the user and his/her avatar can improve the future design of step length modulations aimed at inducing corrections in the user. One potential application for such a method is for gait rehabilitation in post-stroke individuals who suffer from gait asymmetry. Despite recent advances in gait rehab, many stroke patients continue to have residual gait asymmetry [4].

In such a context, the avatar's step length can diminish the appearance of a deficit or inversely amplify it by exaggerating the gait asymmetry. The avatar's modified movements could implicitly affect the patients' movement to alter their step lengths and drive a more symmetrical pattern. Furthermore, the use of modulated self-avatars in the context of post-stroke rehab has been proposed to enhance motor learning as the error between a motor command and its desired outcome can lead to motor adjustments in order to better align the intended motor commands with their perceived result [5].

The purpose of this PhD research is to investigate the effects of modulating an embodied self-avatar's step length on the gait of both healthy participants and stroke patients walking on a treadmill.

## Objectives

The main objective of this study is to develop a self-avatar which is able to create the ownership illusion and whose gait parameters can be modulated in real-time and verify its capability to reduce gait asymmetry in post-stroke patients. The sub-objectives are the following:

1. To develop a tool that is able to apply real-time modulation of gait parameters to an embodied self-avatar.
2. To determine the perception threshold of the step length modulation between user and their self-avatar.
3. To determine if it is possible to induce changes in kinematic and spatiotemporal gait parameters by altering the step length of embodied self-avatar in healthy participants.
4. To test the feasibility of inducing kinematics and spatiotemporal gait parameters by altering the step length of embodied self-avatar in stroke gait.

## 3 METHODS

### 3.1 Materials

All acquisitions involved in this project will be performed on an AMTI split-belt treadmill that measures ground reaction forces. The participants will be fitted with 16 rigid body marker clusters tracked by a 12-camera optoelectronic system running Vicon Tracker 3.6, which measures kinematic data. Vicon Pegasus Advanced 1.2.2 is used for retargeting the body positions onto the virtual avatar. The avatar is generated by Mixamo (Adobe). The VE is animated in Unity 3D game engine and is a replica of our research lab with a large virtual mirror placed in front of the participant. The VE is visualized through a head-mounted display (HMD), the HTC VIVE Pro Eye with a first-person perspective, relative to the avatar. Eye-tracking data will be collected thanks to the HMD.

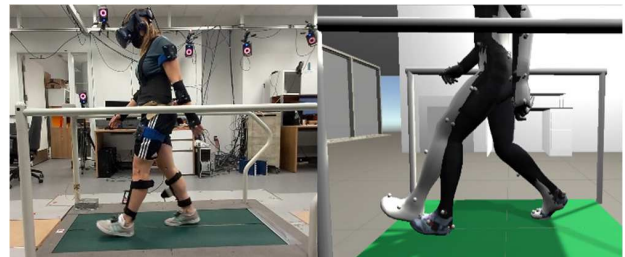


Figure 1: Left: A participant wearing mocap marker clusters and walking on a AMTI split-belt treadmill. Right: Step length modulation illustrated in grey, black leg represents the baseline step length of the participant.

### 3.2 Phase 1: Development of a modulation tool

The project is currently in its first phase, which is dedicated to the development of an algorithm that is able to control the avatar's modulation over time. The algorithm allows the operator to gradually change the step length of one of the avatar legs in real-time. The step length is gradually modulated from mid-swing until the heel strike of the same foot, where the modulation gradually returns to zero. A drastic change of avatar modulation has been

\*iris.willaert.1@ens.etsmtl.ca

shown to lower the embodiment and lowers the follower effect [3]. To preserve the visuomotor synchronization, which has been shown to maintain the ownership illusion, the moment of the new modulated heel strike of the avatar will always correspond with that of the user [6]. The Final IK asset was used to solve the new position of the avatar's kinematics.

### 3.3 Phase 2: Perception threshold of gait symmetry between the user and their modulated self-avatar

Previous results have shown that unconscious motor adaptations can occur within a virtual body ownership paradigm by altering the movements of an embodied avatar [2, 3].

However, it is still unknown to what extent healthy subjects can follow the avatar's movements before the mismatch between actual and modulated movements is detected? Understanding the perception threshold of gait symmetry between subject and avatar could provide a better understanding as to the extent embodied healthy subjects can perceive their avatars as being their own body.

Additionally, it would reveal how large the spatial offset between subject and avatar must be to observe a motor adjustment in order to minimize the sensory conflict.

#### 3.3.1 Proposed protocol

In this phase, 20-30 healthy participants will undergo a protocol where the step length of their avatar will be gradually augmented/reduced over time until the mismatch between real and virtual step length is detected. The avatar's step length will be increased/decreased 1 cm every 10 heel strikes of the same modulated leg until the point of detection is verbally reported. Two different thresholds will be evaluated. The point of detection when the avatar's step length is lengthened (LengtheningThreshold), as well as the point of detection when shortening the avatar's step length, will be obtained (ShorteningThreshold). Each threshold will be evaluated using an ascending/descending series approach. In the ascending series, the avatar's step length will gradually increase the gait asymmetry until the participant reports that he/she perceives that the avatar is not accurately replicating his/her movements. Visa versa for the descending series where the avatar's step length will be decreased until the point of detection is reported. The two thresholds will be evaluated for both legs. This is done as leg dominance may play a role in detecting symmetry between the participant's kinematics and that of the modulated avatar. Participants are not aware of which leg will be modulated during the experimental trials. Kinematic measurement, eye-tracking data will be used to evaluate the perception thresholds. The embodiment questionnaires [7] will be used to assess the strength of the illusion and if a break in illusion occurred at the point of detection. After detection, participants will also be asked what information they used to judge the congruency between themselves and the avatar.

#### 3.3.2 Expected results

We hypothesize that participants will be able to detect the difference between their step length and that of their self-avatar. The ShorteningThreshold will be detected faster and therefore has a lower threshold than the lengthening threshold in order to maintain and maximize their stability during gait. Therefore, the result of phase 1 will be both detection threshold and their effect on the ownership illusion.

### 3.4 Phase 3: Avatar's step length modulation effects on healthy subjects

Here, we will study the effects of modulating the step length of an avatar on 20-30 healthy subjects during treadmill walking. The total offset in terms of step length modulation will be determined in phase 1 to maintain the modulation below the detection threshold.

To attempt to simulate real conditions of asymmetrical gait as seen following a stroke, a weight will be added to one leg of the subjects. This has previously been shown to induce gait asymmetry in healthy subjects [8]. To evaluate if the avatar modulations can restore gait symmetry, four different motor learning strategies will be used in this study: two correcting approaches (lengthening the shorter step and shortening the longer step) and two error-driven approaches (exaggerating the lengthening of the longer step and reducing the shorter step). A positive outcome would suggest that the avatar is capable of restoring gait symmetry in healthy subjects, which may be useful for future research and applications in patient populations where restoring gait symmetry is essential.

### 3.5 Phase 4: Feasibility on stroke patients

In this phase, a study to test the feasibility of restoring gait symmetry through virtual body ownership paradigms on 5 post-stroke individuals will be conducted. The most effective strategy in reducing gait asymmetries in healthy subjects (phase 2) will be used and further explored on post-stroke individuals who suffer from asymmetric gait. Here, the avatar could provide improved visual feedback about step length, closer to the intended movements than the movements that are actually being produced by patients, which may reduce gait asymmetries and improve motor performance.

## 4 CONCLUSION

The phenomenon of ownership illusion through avatars has been widely discussed before. However, the effects of altering the avatar's movements in rehabilitation settings still remain unknown. Here, we present different protocols in an attempt to close the knowledge gap, such as the limits of the motor adjustment that occurs within virtual body ownership paradigms and the effects of restoring gait symmetry in healthy and stroke groups by altering the avatar's step length during treadmill walking. The following points would be valuable to discuss:

- Does the proposed protocol allow us to assess the perception threshold between avatar and subject?
- Are we able to measure whether a break in ownership illusion occurred at point of detection?

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